

Jeff Shen

✉: jshenschool@gmail.com
☎: +1 437-345-1552

🐙: github.com/al-jshen
in: [linkedin.com/in/jeff-shen](https://www.linkedin.com/in/jeff-shen)
🌐: jshen.ca

Education

H.B.Sc., Statistics, Astronomy & Astrophysics, Mathematics 2018-2022
University of Toronto

Technical Skills

Data: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Stan, SQL, NoSQL, R
Other: Rust, BLAS/LAPACK, Docker, bash, vim, git, AWS, SymPy, Astropy, NLTK, L^AT_EX, regex

Research Experience

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| 2021 May - 2021 Aug. | Undergraduate Summer Research Fellowship
<i>Canadian Institute for Theoretical Astrophysics</i>
Stuff here. |
| 2020 Aug. - Present | Undergraduate Research Course
<i>Canadian Institute for Theoretical Astrophysics</i>
Supervised by Prof. Gwendolyn Eadie and Prof. Norman Murray . Study of mass distribution of the Milky Way in a Bayesian framework. Implemented a hierarchical Bayesian model for modelling the mass of galactic halos using an adaptive gradient-based sampling algorithm, increasing computational efficiency by up to two orders of magnitude over existing code. |
| 2020 May - 2020 Aug. | Summer Undergraduate Research Program
<i>Dunlap Institute for Astronomy and Astrophysics, University of Toronto</i>
Supervised by Dr. Allison Man . Study of molecular gas in a strongly-lensed galaxy group at $z = 2.9$. Awarded best poster in student competition. Analyzed ALMA data to infer molecular gas masses in several high-redshift galaxies. Resulted in a first-author publication. |

Honours and Awards

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| 2021 May - 2021 Aug. | Undergraduate Summer Research Fellowship, (\$8396 CAD)
Canadian Institute for Theoretical Astrophysics, University of Toronto |
| 2020 May - 2020 Aug. | Summer Undergraduate Research Program Award (\$9,500 CAD)
Dunlap Institute for Astronomy and Astrophysics, University of Toronto |

Publications

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| 2021 May | Jeff Shen , Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$ ", submitted to <i>ApJ</i> |
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Outreach

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| 2019 - Present | Space Science Journalist (Current in Space segment) , <i>The Star Spot</i> podcast
Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - present. |
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Volunteering

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| 2015 - 2018 | Python Workshop Lead , Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest. |
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Interests

Languages: English (native), French, Chinese, Norwegian (basic)
Other: *nix customization, computer graphics, 3D modelling, running, baking, reading, guitar